GOBI IF1200 Fieldbook

APPN Aerial SOP — Locked revision 1.0

APPN – GOBI IF1200 Fieldbook

For a high-level description of the GOBI sensor package and how it sits alongside the other APPN platforms, see the Platforms Overview.

For the GOBI fieldbook covering the DJI M350, see GOBI M350 Fieldbook.

This fieldbook provides a standardised operational guide for APPN GOBI UAV deployments **on the Inspired Flight IF1200A**, supporting safe flight operations, consistent sensor configuration, and high-integrity data capture. It is intended for trained APPN staff conducting hyperspectral, LiDAR, RGB, and GNSS-INS data acquisitions, and promotes repeatability, transparency, and confidence in downstream data analysis across APPN operations.

> **Important**
>
> **This protocol must be followed for all standard APPN GOBI IF1200 UAV flights.** Adherence to these procedures is essential to ensure operational safety, data integrity, and comparability of datasets across deployments. For any flights that **fall outside standard operating procedures**, detailed records must be kept documenting all deviations, including the specific settings changed, the rationale for those changes, and any anticipated implications for data quality or analysis.

Equipment Checklists

> **Note**
>
> Ensure batteries for all equipment are fully charged before heading to the field. Ensure charging cables are available for necessary equipment.

> **Warning**
>
> Cables and adaptors (particularly ethernet and USB) are known to fail in the field on hot days. Bring spares of all critical cables and adaptors wherever possible.

- ☐ **Aircraft (Inspired Flight IF1200A)**
- ☐ Aircraft batteries
- ☐ Landing gear
- ☐ Landing pad
- ☐ Spare parts
- ☐ Tools
- ☐ Logbook
- ☐ **Radio Control Transmitter / Ground Control Station**
- ☐ **GRYFN Gobi**
- ☐ Gobi system
- ☐ Power cables, chargers
- ☐ Ethernet cable
- ☐ SD card

- ☐ *Capture* polygon files
- ☐ **Ground reference kit**
- ☐ Reflectance calibration panels (11%, 30%, 56%, 82%)
- ☐ Calibration validation panels (20%, 45%)
- ☐ Ground control points and RTK GNSS system
- ☐ 2 × folding tables to elevate panels
- ☐ *If over 50 km from CORS base station*, a portable RTK base station (link to GRYFN gitbook)
- ☐ **Accessories**
- ☐ Safety gear (signage and high-vis vests)
- ☐ Aeronautical radio
- ☐ Field laptop and spare batteries
- ☐ External storage media
- ☐ Water, food, esky, sunscreen, bug spray, first aid kit, etc.
- ☐ Spirit bubble, spirit level (or angle measurement) and measuring tape
- ☐ Portable fan (for cooling GOBI during data offload)
- ☐ External power brick (for charging UAV RC)
- ☐ Anemometer (e.g. Kestrel — *TODO: add link*)

Flight Planning – Inspired Flight IF1200A

> **Warning**
>
> Ensure that you apply for UAV flight approvals for locations and dates of flights well in advance.

1. Using a GPS survey system (Emlid, Trimble…) or a GIS software, create a polygon of the area of interest. Make sure your polygon includes the areas where you will place your calibration panels and GCPs, with an additional 5 m buffer to avoid incomplete data.
2. Save this polygon twice as a KML — once as a *survey* polygon and once as a *capture* polygon.
3. If using QGIS, export the polygon as a KML (in Geometry, select *include z-dimension*, and ensure the CRS is set to WGS 84). Import the *capture* KML into the HPI Polygon Tool and export. This polygon sets the activation of the hyperspectral sensor within Hyperspec3.
4. Import the *survey* polygon into QGroundControl.
5. Using both QGroundControl and the GRYFN flight calculator, determine the speed, altitude, and frame period required to survey the area of interest.

- Do not go below 2 m/s or above 8 m/s for stability reasons (GRYFN). Speeds greater than 8 m/s lead to excessive aircraft pitch and speeds less than 2 m/s accentuate the impacts of wind on aircraft stability and cause a visibly less smooth trajectory.
- Altitude and speed will be tested and recommended from APEx results.
- Ensure the frame period is at a minimum of 20% oversampling, the side overlap is > 40% for the SWIR sensor, and the *turnaround distance* is 2× flight speed (> 3× at > 6 m/s).

6. Ensure flight lines are in the direction of planting (GRYFN).

Pre-Flight

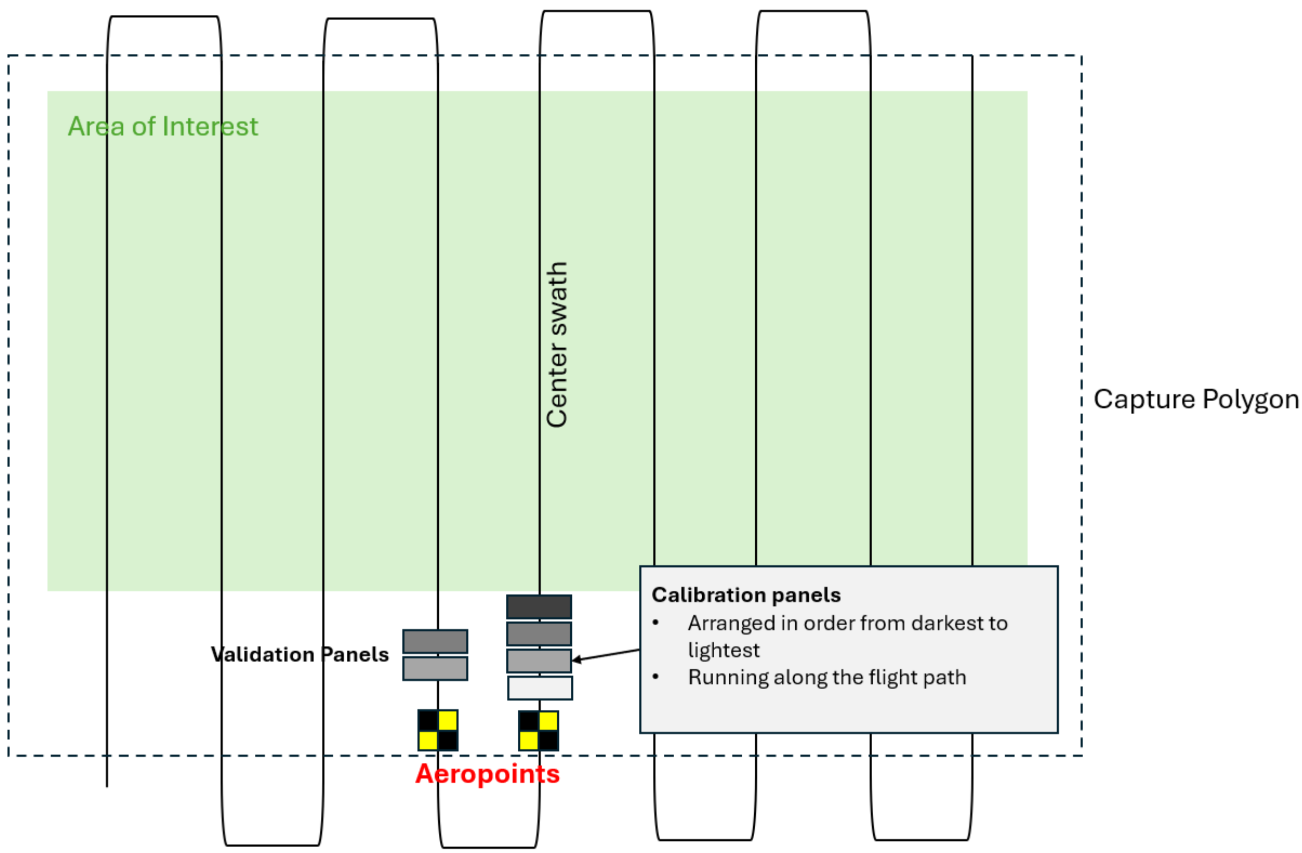
1. Conduct airspace and weather checks.

- No cloud cover.
- Maximum wind speed for quality hyperspectral capture is **6 m/s (~22 km/h)**. Any wind speed over **5 m/s (~18 km/h)** should be recorded.
- Do not operate in conditions below 0 °C or above 40 °C.
- Try to ensure the sun is within ±20° of solar noon (approximately 2 hours before or after noon). **However**, *this depends on time of year and latitude — check NOAA solar calculator if unsure.*

2. Turn on the aeronautical radio and set to local CTAF (find in ERSA).

3. *(Optional)* Set up the Emlid RTK (install a peg or permanent GCP below the base station for recurring flights), let it run for at least 15 minutes, and start recording the RINEX file before flying.
4. Set up reflectance targets for calibration and validation, using a spirit bubble to ensure they are all laid flat. Place 2 GCPs in the field to ensure geometric calibration. GCPs should be located next to calibration panels, ensuring they are in alternate flight lines. (TO DO: ADD STANDARD FLIGHT LAYOUT md)

- Both the calibration and validation panels should be placed on level folding tables to avoid dirt, reduce angle, and reduce spectral spillover. Keeping them level is important.
- 2 sets of GRYFN calibration reflectance panels should be used if available, especially for flight lengths > 15 minutes.
- When using **1** calibration panel, the panel should be placed in the centre of a flight line in the middle of the flight.
- When using **2** panels, they should be placed in the centre of flight lines 1/3 and 2/3 of the mission, noting that all missions would be less than 30 minutes long due to limitations of the IF1200.
- You will need to move the panels if you will be flying multiple missions. Panels need to be in every flight in multi-flight mission capture so that you fly over a calibration panel approximately every 15 minutes.



5. Set up the landing pad and UAV in a safe RTH location.

> **Important**
>
> In dusty environments, an additional tarp must be used under the landing pad.

6. On the controller, set up a safe UAV RTH location, RTH altitude, and other geo-fencing settings on the IF1200.
7. Attach payload to the aircraft:

> **Caution**
>
> **TBC for GOBI on IF1200.** The IF1200 dovetail has no hot-swap protection, so ensure the IF1200 is powered

> off when attaching or removing the sensor (more details).

- Check that all lenses and sensors are clean. If not, use professional lens cleaning wipes (e.g. Zeiss Lens Wipes, recommended by GRYFN) to clean them.
- Connect the power cable from aircraft to payload (non-standard payload bus only).
- Connect both GNSS antenna cables to the correct ports (match A1 and A2).
- Remove RGB and Nano HP lens caps.
- Insert RGB SD card.

8. Power on the radio controller; check battery status.
9. Launch QGroundControl on the IF1200 controller.
10. Review the flight plan, checking operational height and double-checking that area of interest, GCPs, and reflectance panels are all within the capture polygon.
11. Power on the aircraft; confirm connection to GCS and battery status; ensure Remote ID is enabled.
12. The IF1200 does not use RTK; ensure a minimum of 8 satellites and GPS lock before flying.
13. Upload the flight plan to the aircraft.

GOBI Sensor Configuration

1. Place the 80% exposure reference panel under the Nano HP lens; avoid casting shadows.

- Place the drone legs upon exposure-angle cones, ensuring a fixed angle is achieved each exposure setting. **(angle provided through SIF excellence)**
- The maximum distance from the VNIR HS sensor to the GOBI reflectance panel is 40 cm. This is calculated based on the FOV and a 20 cm panel size.

2. Connect to GOBI Wi-Fi or connect the ethernet cable to the payload. *If using ethernet, ensure a static IP address for the sensor at* `10.0.65.2` (or anything other than `50`, `100`, and `128`, and must be less than `255`).
3. Navigate to the GOBI WebUI at `10.0.65.50`.
4. Ensure a valid altitude is shown in the top-right corner of the WebUI (~10–20 m error is acceptable). Typically, this will normalise to ground elevation within 4–6 minutes.
5. Name the mission, using the convention `YYYYMMDD_XXXX` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).
6. Ensure the correct UAV (IF1200) is selected in the dropdown box. **This is very important.**

> **Caution**
>
> Selecting the incorrect drone will render all data captured unrecoverable

7. Scroll down to polygon settings. Ensure triggering is set by **Polygon** in the dropdown menu. Set min and, optionally, max altitude triggers (recommended to set a minimum of at least 20 m to prevent unnecessary data capture).
8. Import the polygon KML or select from previously flown missions using the dropdown menu. Inspect the map to ensure it aligns with expectations.
9. Return to the top of the WebUI, open the Flight Calculator, and update flying height / flight speed; adjust the oversampling buffer (must be at least 20%, recommended setting is 30%). **(value may change through hyperspectral excellence)** Frame period will be automatically updated.
10. Adjust exposure until ~95% spectral saturation at the **peak of the 90th percentile line** without exceeding the frame period. Use the lowest gain mode that still provides sufficient saturation. **(value may change through hyperspectral excellence)**
11. Place the lens cap on the Nano HP.
12. Click **Collect Dark Reference**; wait to finish.
13. **Remove the lens cap.**
14. Remove the exposure reference panel and return it to its case.
15. Press **Start Mission**.
16. Remove the ethernet cable (if using) and begin flight operations.

Flight Operations

1. Ensure the aircraft is in **Position** flight mode.
2. Sync / verify the flight plan in QGroundControl; double-check the flight plan.
3. Clear people/objects away from the UAV.
4. Notify crew/observers that takeoff is beginning.
5. Begin manual takeoff, check stick controls work, and then fly to ~12 m AGL; **do not exceed** the trigger altitude (20 m default) before mission start.
6. Perform dynamic alignment (one to two figure-8 patterns) at a recommended speed of **5 m/s**, at a lower height than capture height and outside of the capture polygon area.
7. Enable autonomous mission.
8. Monitor UAV battery voltage/percentage in flight.
9. After mission, switch back to **Position** mode to regain manual control.
10. Lower to capture altitude and perform post-mission dynamic alignment (figure-8).
11. Land and disarm UAV.
12. Leave the UAV, transmitter, and sensor powered on; begin post-flight checks. Ensure data transfer is finished before turning off the sensor.
13. **Do not hot-swap the batteries.** Treat each flight as a new flight.

Post-Flight Sensor Configuration

1. Reconnect to GOBI Wi-Fi or ethernet cable.
2. Open a browser to `10.0.65.50` or refresh the UI.
3. Press **Stop Mission**.
4. Replace the Nano HP and RGB lens caps.
5. In the WebUI, ensure all data has captured correctly (see the Data Confirmation table below).

Data Confirmation

Sensor type	File count	File size	Notes
GNSS-INS	1 file	~1.5 MB per minute	A new file will be created if time rolls ov
LiDAR	1 per minute	~175 MB per file	First & last file may be smaller. Files ma
VNIR	Mission time (s) / Frame Period value ~ number of data cubes	Several GB even for short flights.	Check that `frameindex`, `imu_gps`, and `sett`
RGB	One image every 2 s* (by default)	~30–70 MB per image	Check the event file vs the number of ima

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Post-Flight

1. Power off the aircraft, then the controller. Inspect the aircraft for any damage and report for maintenance if necessary.
2. Disconnect the ethernet cable from the payload, if using.
3. Disconnect the aircraft batteries.
4. Disconnect the payload power cable.
5. Disconnect the payload GNSS cables.
6. Undo the mounting clamp, remove the payload, and place it in its case.
7. Pack up the aircraft and all gear from site, double-checking against the checklist at the start of this document to ensure everything is accounted for.

Offload Data

1. Open WinSCP, FileZilla, or similar FTP software.
2. Connect to the GRYFN Gobi via `gryfn@10.0.65.50` (password: `gryfn`).

3. Raw GNSS & LiDAR PCAP location: `/data/{mission name}`.
4. VNIR data location: `/data/capturedData/captured/{dateTime}`.
5. GOBI logs location: `/data/gryfn.log.{date}`.
6. RGB images are stored on the camera SD card.
7. Download GNSS, LiDAR, logs, and RGB after each mission; download hyperspectral data later due to its size.
8. Clear data directories after download and backup to avoid filling the 500 GB SSD.

Data storage, processing & validation

All paths below follow the APPN folder structure. Formal paths use the wiki's placeholder syntax; an example follows each.

1. Downloaded data should be stored in the correct `T0_raw` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/
```

2. Data from the GCP points should be saved in the `T0_raw/Vault` folder (location may change).

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/Vault/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/Vault/
```

3. Important information and issues should be recorded in the `FieldNotes.txt` file.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/FieldNotes.txt
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/FieldNotes.txt
```

4. Per-run information (e.g. APEx test conditions) should be stored in `RunOverview.csv`.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/RunOverview.csv
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/RunOverview.csv
```

> **Important**
>
> When data from a failed run is being kept (e.g. debugging with GRYFN), the `RunFailed` boolean column of `RunOverview.csv` must be set to `True`.

5. The bundled `.graw` should be saved in the same `T0_raw` folder.
6. The `.gpro` should be generated using the APPN GPT Pipeline (`.json` and wiki links *TODO: add links*) and stored in the adjacent `T1_proc` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T1_proc/
```

7. Standard QA process should be performed following this guide.

Appendix

FTP & Service Reference

Service	Protocol	IP	Port	User	Password
GOBI	FTP	`10.0.65.50`	22	`gryfn`	`gryfn`
SBG	FTP	`10.0.65.100`	21	`operator`	*(none)*
Headwall polygon	HTTP	`apps.headwallphotonics.com`	—	—	—

Headwall Polygon Tool — Workflow

1. Import KML.

- Must be in Google Earth format.

2. Export KML.
3. Save KML.
4. In File Explorer, rename the KML as the survey name.

Setting a Static IP Address

If connecting over an ethernet connection, a static IP address will need to be set:

1. Open *Network and Internet Settings*.
2. Open *Change Adapter Options*.
3. Open *Properties* for the Ethernet port/adapter.
4. Open *Properties* for IPv4.
5. Set IP address to `10.0.65.2`.
6. Set subnet mask to `255.255.255.0`.
7. Press **OK**.

IP Address Reference

Service	Address
GRYFN WebUI	`10.0.65.50`
HSInsight	`10.0.65.50:8080`
Ouster	`10.0.65.128`
SBG	`10.0.65.100`